

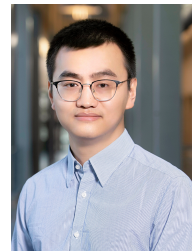
Jilin Hu

Full Professor, School of Data Science and Engineering, East China Normal University. His work focuses on spatio-temporal data management, traffic analytics, time series, and AI for science.

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Education and Employment

- 2023 – present 📌 **Professor, Zijiang Young Scholar**, School of Data Science and Engineering, East China Normal University.
- 2022 – 2023 📌 **Associate Professor**, Department of Computer Science, Aalborg University, Denmark.
- 2020 – 2022 📌 **Assistant Professor**, Department of Computer Science, Aalborg University, Denmark.
- 2019 – 2020 📌 **Research Scientist**, Inception Institute of Artificial Intelligence, United Arab Emirates.
- 2017 – 2018 📌 **Visiting PhD Student**, Intelligent Transportation Research Group, University of California, Berkeley, USA.
- 2015 – 2019 📌 **PhD Student**, Department of Computer Science, Aalborg University, Denmark.
- 2011 – 2014 📌 **MSc Student**, School of Aeronautics and Astronautics, Shanghai Jiao Tong University, China.
- 2007 – 2011 📌 **BSc Student**, School of Metrology and Testing Engineering, China Jiliang University, China.

Research Grants

- 2025 – 2028 📌 **National Natural Science Foundation of China**, General Program, Principal Investigator, Research on Multimodal Universal Pretraining for Trajectory Analytics, RMB 500k.
- 2023 – 2026 📌 **National Natural Science Foundation of China**, Excellent Young Scientists Fund (Overseas), Principal Investigator, Spatio-temporal Data Management and Analytics, RMB 1M.
- 2024 – 2026 📌 **Villum Foundation**, Synergy, Co-Principal Investigator, Inverse Design of Materials Using Diffusion Probabilistic Models, DKK 2.99M.
- 2023 – 2026 📌 **East China Normal University Basic Research Funds**, Key Research Initiative, Principal Investigator, Personalized Deposit Rate Pricing for Banking Customers, RMB 500k.

Honors and Awards

- 2025 📌 **EMNLP 2025 Best Paper (Industrial) Award.**
- 2024 📌 **VLDB 2024 Best Paper Award Nomination.**
- 2023 📌 **ACM SIGSPATIAL China Rising Star Award.**
- 2022 📌 **ICDE 2022 Best Paper Award.**
- 📌 **Selected for the National Youth Talent Program.**

Professional Service

- 2024 – present ■ **Executive Committee Member, CCF Large Model Forum.**
 ■ **Member, Intelligent and Digital Surgery Committee, Shanghai Medical Association.**
- 2023 – present ■ **Executive Committee Member, CCF Database Technical Committee.**
 ■ **Member, CAAI Intelligent Service Technical Committee.**
- 2022 – present ■ **Chair, Denmark Chapter, IEEE Computer Society.**
 ■ **Executive Committee Member, ACM SIGSPATIAL China Chapter.**

Publications

Journal Articles

- [1] Jonas A. Finkler, Yan Lin, Tao Du, **Jilin Hu**, and Morten M. Smedskjaer. "Inverse Design of Amorphous Materials With Targeted Properties". In: *Advanced Materials* (2026), e22493.
- [2] Shiyan Hu, Kai Zhao, Chenjuan Guo, Xiangfei Qiu, Yang Shu, **Jilin Hu**, Christian S. Jensen, and Bin Yang. "TAP: Time Series Anomaly Prediction Via Adaptive Period Modeling and Dual Representation Learning". In: *IEEE Transactions on Knowledge and Data Engineering* (2026), pp. 1–14.
- [3] Haomin Yu, **Jilin Hu**, Yunyao Cheng, Chenjuan Guo, Yizhou Zhu, Bin Yang, and Christian S. Jensen. "A Knowledge-Based Semi-Supervised Crystal Property Prediction Framework with Consistency Regularization". In: *IEEE Transactions on Knowledge and Data Engineering* (2026), pp. 1–13.
- [4] Haomin Yu, **Jilin Hu**, Kasper Tolborg, Chenjuan Guo, Mohamed Saraee, Bin Yang, and Christian S. Jensen. "A Selective Multi-View Representation Augmentation Framework for Crystal Property Prediction". In: *ACM Transactions on AI for Science* (2026).
- [5] Kai Zhang, Chaoxiang Lan, Yazhang Xu, Zheyang Li, Wenming Tan, Ye Ren, and **Jilin Hu**. "SparseLight: Dynamic gradient-optimized softmax for efficient transformer acceleration". In: *Knowledge-Based Systems* 333 (2026), p. 115050.
- [6] Letian Gong, Shengnan Guo, Yan Lin, Yichen Liu, Erwen Zheng, Yiwei Shuang, Youfang Lin, **Jilin Hu**, and Huaiyu Wan. "STCDM: Spatio-Temporal Contrastive Diffusion Model for Check-In Sequence Generation". In: *IEEE Trans. Knowl. Data Eng.* 37.4 (2025), pp. 2141–2154.
- [7] Yan Lin, **Jilin Hu**, Shengnan Guo, Bin Yang, Christian S. Jensen, Youfang Lin, and Huaiyu Wan. "UVTM: Universal Vehicle Trajectory Modeling with ST Feature Domain Generation". In: *IEEE Trans. Knowl. Data Eng.* (2025).
- [8] Yan Lin, Huaiyu Wan, Shengnan Guo, **Jilin Hu**, Christian S. Jensen, and Youfang Lin. "Pre-Training General Trajectory Embeddings With Maximum Multi-View Entropy Coding". In: *IEEE Trans. Knowl. Data Eng.* 36.12 (2024), pp. 9037–9050.
- [9] Di Qi, **Jilin Hu**, and Jianbing Shen. "Few-Shot Object Detection With Self-Supervising and Cooperative Classifier". In: *IEEE TNNLS* 35.4 (2024), pp. 5435–5446.
- [10] Xinle Wu, Xingjian Wu, Bin Yang, Lekui Zhou, Chenjuan Guo, Xiangfei Qiu, **Jilin Hu**, Zhenli Sheng, and Christian S. Jensen. "AutoCTS++: zero-shot joint neural architecture and hyperparameter search for correlated time series forecasting". In: *VLDB J.* 33.5 (2024), pp. 1743–1770.
- [11] Xinle Wu, Xingjian Wu, Bin Yang, Lekui Zhou, Chenjuan Guo, Xiangfei Qiu, **Jilin Hu**, Zhenli Sheng, and Christian S. Jensen. "AutoCTS++: zero-shot joint neural architecture

- and hyperparameter search for correlated time series forecasting". In: *VLDB J.* 33.5 (2024), pp. 1743–1770.
- [12] Simin Xu, Lingkun Luo, Haichao Hong, **Jilin Hu**, Bin Yang, and Shiqiang Hu. "Multi-granularity attention in attention for person re-identification in aerial images". In: *Vis. Comput.* 40.6 (2024), pp. 4149–4166.
 - [13] Haomin Yu, **Jilin Hu**, Xinyuan Zhou, Chenjuan Guo, Bin Yang, and Qingyong Li. "CGF: A Category Guidance Based PM_{2.5} Sequence Forecasting Training Framework". In: *IEEE TKDE* (2023), pp. 1–14.
 - [14] Simin Xu, Lingkun Luo, **Jilin Hu**, Bin Yang, and Shiqiang Hu. "Semantic driven attention network with attribute learning for unsupervised person re-identification". In: *Knowledge-Based Systems* 252 (2022), p. 109354.
 - [15] Simin Xu, Lingkun Luo, **Jilin Hu**, Bin Yang, and Shiqiang Hu. "Semantic driven attention network with attribute learning for unsupervised person re-identification". In: *Knowledge-Based Systems* (2022), p. 109354.
 - [16] Huanlong Zhang, Jiapeng Zhang, Guohao Nie, **Jilin Hu**, and W.J. (Chris) Zhang. "Residual memory inference network for regression tracking with weighted gradient harmonized loss". In: *Information Sciences* 597 (2022), pp. 105–124.
 - [17] Jing Li, Peng Han, Xiangnan Ren, **Jilin Hu**, Lisi Chen, and Shuo Shang. "Sequence Labeling with Meta-Learning". In: *IEEE Transactions on Knowledge and Data Engineering* (2021), pp. 1–1.
 - [18] Yong Wang, Xian Wei, Hao Shen, **Jilin Hu**, and Lingkun Luo. "Performance evaluation of low resolution visual tracking for unmanned aerial vehicles". In: *Neural Comput. Appl.* 33.7 (2021), pp. 2229–2248.
 - [19] Bolong Zheng, Qi Hu, Lingfeng Ming, **Jilin Hu**, Lu Chen, Kai Zheng, and Christian S. Jensen. "SOUP: Spatial-Temporal Demand Forecasting and Competitive Supply". In: *IEEE TKDE* (2021), pp. 1–1.
 - [20] Chenjuan Guo, Bin Yang, **Jilin Hu**, Christian S. Jensen, and Lu Chen. "Context-aware, preference-based vehicle routing". In: *VLDB J.* 29.5 (2020), pp. 1149–1170.
 - [21] **Jilin Hu**, Bin Yang, Chenjuan Guo, and Christian S. Jensen. "Risk-aware path selection with time-varying, uncertain travel costs: a time series approach". In: *VLDB J.* 27.2 (2018), pp. 179–200.
 - [22] Bin Yang, Jian Dai, Chenjuan Guo, Christian S. Jensen, and **Jilin Hu**. "PACE: a PATH-Centric paradigm for stochastic path finding". In: *VLDB J.* 27.2 (2018), pp. 153–178.
 - [23] **Jilin Hu**, Bin Yang, Christian S. Jensen, and Yu Ma. "Enabling time-dependent uncertain eco-weights for road networks". In: *GeoInformatica* 21.1 (2017), pp. 57–88.

Conference Papers

- [1] Yujie Li, Zezhi Shao, Chengqing Yu, Yisong Fu, Weijie Zhu, Yifan Du, **Jilin Hu**, Bin Yang, Yongjun Xu, and Fei Wang. "QUALS: Corpus Equilibrium for Universal Forecasting via Pattern Quantization and Learnability Synchronization". In: *PVLDB*. 2027.
- [2] Chaofan Wang, Ziqiang Yu, Xiaohui Yu, **Jilin Hu**, Meng Chen, and Yang Liu. "CRSTNet: Efficient Spatio-Temporal Forecasting via Adaptive Cluster Routing". In: *ICDE*. 2027.
- [3] Sean Bin Yang, Ying Sun, Zongyi Xu, Tung Kieu, **Jilin Hu**, Bin Yang, Kristian Torp, Hua Lu, and Torben Bach Pedersen. "When Correlations Mislead: Confounder-Aware Multi-View Urban Region Representation Learning". In: *ICDE*. 2027.

- [4] Yuhao Zhu, **Jilin Hu**, Xinying Cai, Yingshan Li, Li Ma, Xiangfei Qiu, Linsen Li, Kai Zhang, Yao Fu, Weihao Jiang, and Bin Yang. "WPBench: A Comprehensive Benchmark for Wind Power Forecasting". In: *ICDE*. 2027.
- [5] Yongru Chen, Kai Zhang, Zeliang Zong, Yuchen Lu, Wenming Tan, Ye Ren, and **Jilin Hu**. "One Layer's Trash is Another Layer's Treasure: Adaptive Layer-wise Visual Token Selection in LVLMS". In: *CVPR*. 2026.
- [6] Hongfan Gao, Wangmeng Shen, Bin Yang, and **Jilin Hu**. "HyFAD: Hybrid Time-Frequency Diffusion with Frequency-Aware Embedding for Time Series Imputation". In: *NeurIPS*. 2026.
- [7] Chenxu Han, Sean Bin Yang, and **Jilin Hu**. "DiffMM: Efficient Method for Accurate Noisy and Sparse Trajectory Map Matching via One Step Diffusion". In: *AAAI*. 2026.
- [8] Zhengyu Li, Xiangfei Qiu, Yuhao Zhu, Xingjian Wu, **Jilin Hu**, Chenjuan Guo, and Bin Yang. "GCGNet: Graph-Consistent Generative Network for Time Series Forecasting with Exogenous Variables". In: *ICLR*. 2026.
- [9] Yan Lin, **Jilin Hu**, Shengnan Guo, Christian S. Jensen, Youfang Lin, and Huaiyu Wan. "DiSGMM: A Method for Time-varying Microscopic Weight Completion on Road Networks". In: *IJCAI*. 2026.
- [10] Yan Lin, **Jilin Hu**, N. M. Anoop Krishnan, and Morten M. Smedskjaer. "AMGenC: Generating Charge Balanced Amorphous Materials". In: *NeurIPS*. 2026.
- [11] Xvyuan Liu, Xiangfei Qiu, Hanyin Cheng, Xingjian Wu, Chenjuan Guo, Bin Yang, and **Jilin Hu**. "ASTGI: Adaptive Spatio-Temporal Graph Interactions for Irregular Multivariate Time Series Forecasting". In: *ICLR*. 2026.
- [12] Xvyuan Liu, Xiangfei Qiu, Xingjian Wu, Zhengyu Li, Chenjuan Guo, **Jilin Hu**, and Bin Yang. "Rethinking Irregular Time Series Forecasting: A Simple yet Effective Baseline". In: *AAAI*. 2026.
- [13] Jiahong Lyu, Hongfan Gao, Wangmeng Shen, Qingsong Zhong, and **Jilin Hu**. "PolarFormer: Radial-Angular Latent Modeling for Unconditional Time Series Generation". In: *KDD*. 2026.
- [14] Xiaowei Mao, Huihu Ding, Yan Lin, Tingrui Wu, Shengnan Guo, Dazhuo Qiu, Feiling Fang, **Jilin Hu**, and Huaiyu Wan. "Spatial-Temporal Feedback Diffusion Guidance for Controlled Traffic Imputation". In: *AAAI*. 2026.
- [15] Xiangfei Qiu, Xvyuan Liu, Tianen Shen, Xingjian Wu, Hanyin Cheng, Bin Yang, and **Jilin Hu**. "SEER: Transformer-based Robust Time Series Forecasting via Automated Patch Enhancement and Replacement". In: *ICML*. 2026.
- [16] Xiangfei Qiu, Kangjia Yan, Xvyuan Liu, Xingjian Wu, and **Jilin Hu**. "Bridging Time and Frequency: A Joint Modeling Framework for Irregular Multivariate Time Series Forecasting". In: *ICML*. 2026.
- [17] Xiangfei Qiu, Liu Yang, Xiangyu Xu, Hanyin Cheng, Xingjian Wu, Rongjia Wu, Zhigang Zhang, Ding Tu, Chenjuan Guo, Bin Yang, Christian S. Jensen, and **Jilin Hu**. "Hermes: A Multi-Scale Spatial-Temporal Hypergraph Network for Stock Time Series Forecasting". In: *NeurIPS*. 2026.
- [18] Xiangfei Qiu, Yuhao Zhu, Zhengyu Li, Xingjian Wu, Bin Yang, and **Jilin Hu**. "DAG: A Dual Correlation Network for Time Series Forecasting with Exogenous Variables". In: *ICML*. 2026.
- [19] Wangmeng Shen, Hongfan Gao, Qingsong Zhong, Dingli Xu, and **Jilin Hu**. "FSDI: Frequency-Shaped Diffusion For Time-Series Imputation". In: *The Web Conference*. 2026.

- [20] Xingjian Wu, Hanyin Cheng, Xiangfei Qiu, Zhengyu Li, **Jilin Hu**, Chenjuan Guo, and Bin Yang. "FLAME: Flow Enhanced Legendre Memory Models for General Time Series Forecasting". In: *NeurIPS*. 2026.
- [21] Xingjian Wu, Xvyuan Liu, Junkai Lu, Siyuan Wang, Xiangfei Qiu, Yang Shu, **Jilin Hu**, Chenjuan Guo, and Bin Yang. "ST-EVO: Towards Generative Spatio-Temporal Evolution of Multi-Agent Communication Topologies". In: *EMNLP*. 2026.
- [22] Xiangyu Xu, Qingsong Zhong, and **Jilin Hu**. "TimeMar: Multi-Scale Autoregressive Modeling for Unconditional Time Series Generation". In: *The Web Conference*. 2026.
- [23] Sean Bin Yang, Hao Miao, Zongyi Xu, **Jilin Hu**, Xiangmeng Wang, Hua Lu, Bin Yang, and Christian S. Jensen. "DGCPATH: Distribution-Aware Generative Contrastive Framework for Self-supervised Path Representation Learning". In: *IJCAI*. 2026.
- [24] Sean Bin Yang, Ying Sun, **Jilin Hu**, Zongyi Xu, Kristian Torp, Hua Lu, Bin Yang, and Christian S. Jensen. "REFINE: Trajectory Representation Learning via Closed-Loop Transcription". In: *KDD*. 2026.
- [25] Xiaoyun Yu, Li Fan, Xiangfei Qiu, Nanqing Dong, Yonggui Huang, Honggang Qi, Geguang Pu, Wanli Ouyang, Xi Chen, and **Jilin Hu**. "MEMTS: Internalizing Domain Knowledge via Parameterized Memory for Retrieval-Free Domain Adaptation of Time Series Foundation Models". In: *KDD*. 2026.
- [26] Buang Zhang, Tung Kieu, Xiangfei Qiu, Chenjuan Guo, **Jilin Hu**, Aoying Zhou, Christian S. Jensen, and Bin Yang. "An Encode-then-Decompose Approach to Unsupervised Time Series Anomaly Detection on Contaminated Training Data". In: *ICDE*. 2026.
- [27] Qingsong Zhong, Haomin Yu, Yan Lin, Wangmeng Shen, Long Zeng, and **Jilin Hu**. "SculptDrug: A Spatial Condition-Aware Bayesian Flow Model for Structure-based Drug Design". In: *AAAI*. 2026.
- [28] Zeliang Zong, Kai Zhang, Yarong Wang, Wenming Tan, Ye Ren, and **Jilin Hu**. "Less Token, More Signal: MoE Expert Pruning via Critical Token Selection". In: *ICML*. 2026.
- [29] Hongfan Gao, Wangmeng Shen, Xiangfei Qiu, Ronghui Xu, Bin Yang, and **Jilin Hu**. "SSD-TS: Exploring the potential of linear state space models for diffusion models in time series imputation". In: *KDD*. 2025.
- [30] Zhe Li, Xiangfei Qiu, Peng Chen, Yihang Wang, Hanyin Cheng, Yang Shu, **Jilin Hu**, Chenjuan Guo, Aoying Zhou, Christian S. Jensen, and Bin Yang. "FM4TS-Bench: A Comprehensive and Unified Benchmark of Foundation Models for Time Series Forecasting". In: *KDD*. 2025.
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- [32] Xiangfei Qiu, Zhe Li, Wanghui Qiu, Shiyan Hu, Lekui Zhou, Xingjian Wu, Zhengyu Li, Chenjuan Guo, Aoying Zhou, Zhenli Sheng, **Jilin Hu**, Christian S. Jensen, and Bin Yang. "TAB: Unified Benchmarking of Time Series Anomaly Detection Methods". In: *PVLDB*. 2025.
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Flavio Chierichetti, Hady W. Lauw, Claudia Perlich, Wee Hyong Tok, and Andrew Tomkins. ACM, 2025, pp. 1185–1196.

- [35] Haichen Wang, Liu Yang, Xinyuan Zhang, Haomin Yu, Ming Li, and **Jilin Hu**. "ADFormer: Aggregation Differential Transformer for Passenger Demand Forecasting". In: *IJCAI*. 2025.
- [36] Tonglong Wei, Yan Lin, Youfang Lin, Shengnan Guo, **Jilin Hu**, Haitao Yuan, Gao Cong, and Huaiyu Wan. "PLMTrajRec: A Scalable and Generalizable Trajectory Recovery Method with Pre-trained Language Models". In: *NeurIPS*. 2025.
- [37] Tonglong Wei, Yan Lin, Zeyu Zhou, Haomin Wen, **Jilin Hu**, Shengnan Guo, Youfang Lin, Gao Cong, and Huaiyu Wan. "TransferTraj: A Vehicle Trajectory Learning Model for Region and Task Transferability". In: *NeurIPS*. 2025.
- [38] Yongfu Wei, Yan Lin, Hongfan Gao, Ronghui Xu, Sean Bin Yang, and **Jilin Hu**. "Path-LLM: A Multi-Modal Path Representation Learning by Aligning and Fusing with Large Language Models". In: *WWW*. Ed. by Guodong Long, Michale Blumstein, Yi Chang, Liane Lewin-Eytan, Zi Helen Huang, and Elad Yom-Tov. ACM, 2025, pp. 2289–2298.
- [39] Wenyu Wu, Jiali Mao, Jiafan Liu, Yixiao Tong, Lisheng Zhao, Shaosheng Cao, **Jilin Hu**, Aoying Zhou, and Lin Zhou. "CDMap: Complementarity and Disparity-aware Map Inference Quality Enhancement". In: *ICDE*. 2025.
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- [43] Ziyang Yang, Shuhui Gong, Xinqi Liu, Jiahao Lv, Changjian Liu, **Jilin Hu**, and Hongbin Pei. "VisitFrequency-Diffusion: Leveraging Recurrent Visits for Long-Term Individual Trajectory Forecasting". In: *SIGSPATIAL*. 2025, pp. 1190–1193.
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- [48] Zhe Li, Ronghui Xu, **Jilin Hu**, Zhong Peng, Xi Lu, Chenjuan Guo, and Bin Yang. "Ocean Significant Wave Height Estimation with Spatio-temporally Aware Large Language Models". In: *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*. Ed. by Edoardo Serra and Francesca Spezzano. 2024, pp. 3892–3896.

- [49] Yan Lin, Huaiyu Wan, **Jilin Hu**, Shengnan Guo, Bin Yang, Youfang Lin, and Christian S. Jensen. "Origin-Destination Travel Time Oracle for Map-based Services". In: *SIGMOD*. 2024.
- [50] Minbo Ma, **Jilin Hu**, Christian S. Jensen, Fei Teng, Peng Han, Zhiqiang Xu, and Tianrui Li. "Learning Time-Aware Graph Structures for Spatially Correlated Time Series Forecasting". In: *ICDE*. 2024, pp. 4435–4448.
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- [54] Tianyi Li, Lu Chen, Christian S. Jensen, Torben Bach Pedersen, and **Jilin Hu**. "Evolutionary Clustering of Streaming Trajectories". In: *ICDE*. 2022.
- [55] Frederik Mathiesen, Bin Yang, and **Jilin Hu**. "Hyperverlet: A Symplectic Hypersolver for Hamiltonian Systems". In: *AAAI*. 2022.
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